INTERIM FEDERAL SPECIFICATION

COATING, CLEAR (SATIN-FINISH), POLYURETHANE, LATEX, FOR INTERIOR USE

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers one type of a water-based one-component polyurethane latex coating, ready for use on floors, walls, ceilings, and furniture.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-S-626 - Sponges, Synthetic.
PPP-P-1892 - Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of.
PPP-T-60 - Tape, Packaging, Waterproof.

Federal Standard:

Fed. Test Method Std. No. 141/GEN - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

D 93 - Flash Point by Pensky-Martens Closed Tester.
D 1475 - Density of Paint, Varnish, Lacquer, and Related Products.
D 1545 - Viscosity of Transparent Liquids by Bubble Time Method.
D 2243 - Freeze Resistance of Latex and Emulsion Paints.
D 2369 - Volatile Content of Paints.
D 2486 - Scrub Resistance of Interior Latex Flat Wall Paints.
D 3273 - Test for Resistance to Growth of Mold on the Surface of Interior Coatings.
D 3335 - Determination of Low Concentrations of Lead by Atomic Absorption Spectroscopy.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The coating under this specification shall be a water dispersion of a high molecular weight urethane polymer. It shall dry at room temperature and provide a continuous film with a satin finish.

3.2 Odor. The odor of the coating shall not be putrid or otherwise offensive or irritating. After application, there shall be no residual odor after 24 hours of drying.

3.3 Qualitative requirements.

3.3.1 Condition in container. A freshly opened container of the coating, when tested as in 4.3.1, shall show no deterioration, skinning, livering, thickening, seeding, or gelling. Any undispersed material shall be readily dispersed with spatula to a homogeneous state.

3.3.2 Storage stability.

3.3.2.1 Partially full container. When tested as specified in 4.3.2, the coating shall not skin within 48 hours.

3.3.2.2 Full container. When tested as specified in 4.3.2, the coating shall show no skinning, livering, curdling, seeding, hard and dry caking, or tough gummy sediment. After storage for 12 months as specified in 4.3.2, it shall remix readily to a homogeneous state, and there shall be no change in the drying time.

3.3.3 Application properties. The coating shall be easily applied by brush and spray equipment, when tested as specified in 4.3.3. The coating shall show no streaks or conspicuous brush marks.

3.3.4 Appearance of dried coating. The dried coating, applied as specified in 4.3.4, shall be smooth, uniform, free from craters and from other defects caused by bubble retention.

3.3.5 Anchorage. The film of the coating, when tested as specified in 4.3.5, shall show no removal or other loosening of the coating beyond 1.6 mm (0.0625 inch) on either side of the score line.

3.3.6 Flexibility. When tested as specified in 4.3.6, the coating shall show no evidence of cracking, chipping, or flaking.

3.3.7 Knife test. A film of the coating, when tested as specified in 4.3.7, shall ribbon or curl without flaking or otherwise loosening from the panel.

3.3.8 Scrub resistance. When coated panels are tested as specified in 4.3.8, the film shall not be worn through to the substrate in less than 500 cycles (1000 separate strokes).

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3.3.9 Impact resistance. A film of the coating, when tested as specified in 4.3.9, shall withstand a minimum impact of 160 inch-pounds without flaking, cracking, chipping, or peeling.

3.3.10 Fungus resistance. The coating shall show no fungus growth when tested as specified in 4.3.10.

3.3.11 Freeze-thaw resistance. The coating shall withstand the freeze-thaw test described in 4.3.11 with a change in consistency not greater than 10 K.U. After completion of the test, the coating applied to a clean plate glass panel shall dry to a smooth, uniform finish.

3.4 Quantitative requirements.

3.4.1 The quantitative requirements of the coating shall be as specified in table I.

TABLE I. Quantitative requirements

Characteristics	Requirements	
	Minimum	Maximum
Total solids, percent by weight of coating	48	50
Flash point, °C (°F)	29 (85)	--
Weight per gallon, lb.	8.9	--
Viscosity, cps	700	1500
Drying time, hour	1/2	--
Abrasion resistance, wear index	--	30
Hardness, Shore A2, Pencil	75	85
Lead content, percent by weight of nonvolatile	--	0.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Sampling and inspection shall be in accordance with Section 1000 of Fed. Test Method Std. No. 141.

4.3 Test procedures. The test procedures shall be conducted in accordance with Fed. Test Method Std. No. 141, except as otherwise specified in table II. Referee test conditions are specified in Section 7 of Fed. Test Method Std. 141.

TABLE II. Index

Characteristics	Requirement reference	Applicable Tests		
		Fed. Test Method Std. No. 141	ASTM	Paragraph reference
Total solids	Table I	----	D2369	----
Flash point	Table I	----	D93	----
Weight per gallon	Table I	----	D1475	----
Viscosity	Table I	----	D1545	----
Drying time	Table I	4061	----	----
Abrasion resistance	Table I	6192	----	4.3.13
Hardness	Table I	----	----	----
Lead content	Table I	----	D3335	4.3.12
Condition in container	3.3.1	3011	----	4.3.1
Storage stability	3.3.2	3021,3022	----	4.3.2
Application properties	3.3.3	----	----	4.3.3

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TABLE II. Index (cont.)

Characteristics	Requirement reference	Applicable Tests: Fed. Test Method Std. No. 141	ASTM	Paragraph reference
Appearance	3.3.4	----	----	4.3.4
Anchorage	3.3.5	----	----	4.3.5
Flexibility	3.3.6	6221	----	4.3.6
Knife test	3.3.7	6304	----	4.3.7
Scrub resistance	3.3.8	----	D2486	4.3.8
Impact resistance	3.3.9	----	----	4.3.9
Fungus resistance	3.3.10	----	D3273	4.3.10
Freeze-thaw resistance	3.3.11	----	D2243	4.3.11

4.3.1 Condition in container. Examine the condition of the coating in the container as received in accordance with method 3011 of Fed. Test Method Std. No. 141, and evaluate for compliance with 3.3.1.

4.3.2 Storage stability. Determine storage stability of the coating in accordance with methods 3021 and 3022 of Fed. Test Method Std. No. 141, and evaluate for compliance with 3.3.2.

4.3.3 Application properties. Prepare two steel panels in accordance with method 2011, paragraph 4.1.1, of Fed. Test Method Std. No. 141. On one panel brush a coat of the coating at a spreading rate of approximately 400 square feet per gallon, and on the other panel spray a coat at approximately the same spreading rate. Allow the panels to dry almost on a vertical position for 24 hours at room temperature. Evaluate for compliance with 3.3.3.

4.3.4 Appearance. Evaluate the dried films prepared in 4.3.3 for compliance with 3.3.4.

4.3.5 Anchorage. Spray a film of the coating to a dry film thickness of 0.025 ± 0.003 mm (0.0010 ± 0.001 inch) on a primed aluminum panel prepared in accordance with method 2013 of Fed. Test Method Std. No. 141. Use a vinyl butyrate wash primer or any other suitable primer, provided it is compatible with the polyurethane coating. Air dry in a horizontal position for 18 hours, then bake at 105 ± 2°C for two hours. Condition the panel for one hour under referee testing conditions, and then score a line through to the metal across the width of the film using a sharp pointed knife. Tape the film perpendicular to and across the score line with waterproof, pressure-sensitive (19 mm wide) tape conforming to PPP-T-60, type IV. The tape shall be pressed using the thumb in firm contact with the film, and shall extend approximately one inch on each side of the score line. All air bubbles shall be rolled out by firm thumb pressure. Grasp a free end of the tape, and at rapid speed strip it from the film by pulling the tape back upon itself at an angle of 180°. Evaluate for compliance with 3.3.5.

4.3.6 Flexibility. Determine flexibility in accordance with method 6221 of Fed. Test Method Std. No. 141. Draw down a 5.1-cm (2-inch) wide film of the coating using a doctor blade that will deposit a dry-film thickness of 0.051 mm (0.002 inch) on a solvent-cleaned steel panel prepared in accordance with method 2011, paragraph 4.1.1, of Fed. Test Method Std. No. 141. The coating shall be allowed to air dry for 7 days to ensure complete curing. Bend over a 3.17 mm (18 inch) mandrel and evaluate for compliance with 3.3.6.

4.3.7 Knife test. Perform the knife test in accordance with method 6304 of Fed. Test Method Std. No. 141, using a flat portion of the panel used in 4.3.9, and evaluate for compliance with 3.3.7.

4.3.8 Scrub resistance. Determine the scrub resistance of the coating using ASTM method D 2486, except use a synthetic sponge conforming to L-S-626, type II, in lieu of a bristle brush. Evaluate for compliance with 3.3.8.

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4.3.9 Impact resistance. Draw down a film of the coating on a 20-gauge cold rolled steel panel, prepared according to method 2011, paragraph 4.1.1, of Fed. Test Method Std. No. 141, to a wet film thickness of 0.076 mm (0.003 inch), and allow to dry at room temperature for two weeks. Subject the coated panel to a reverse impact test using an 80 inch-pound Gardner (or equivalent) variable impact tester. Evaluate for compliance with 3.3.9.

4.3.10 Fungus or mold resistance. Determine the fungus resistance of the coating in accordance with ASTM method D 3273, and evaluate for compliance with 3.3.10.

4.3.11 Freeze-thaw resistance. Determine the freeze-thaw resistance of the coating in accordance with ASTM method D 2243, except that the sample shall not be exposed to temperatures below 0°C (32°F). Evaluate for compliance with 3.3.11.

4.3.12 Lead content. Lead content shall be determined in accordance with ASTM method D 3335, and evaluate for compliance with table I.

4.3.13 Abrasion resistance. Determine the wear index of the coating in accordance with method 6192 of Fed. Test Method Std. No. 141. Draw down a film of the coating with an applicator that will produce a dry-film thickness of 0.025 mm (0.001 inch) on a steel panel prepared according to method 2011, paragraph 4.1.1, of Fed. Test Method Std. 141, and allow to dry at standard conditions for 14 days. Run the abrasion test using a Taber Abraser Model 174, CS-17 Calibrase Wheel, 1000 grams, and 1000 cycles. Evaluate the wear index for compliance with Table I.

4.3.14 Inspection of preparation for delivery. The packaging, packing, and marking of the containers shall be examined and tested to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The coating shall be packaged, packed and marked in accordance with PPP-P-1892. The level of packaging shall be A, B, or C, and the level of packing shall be A, B, or C, as specified (see 6.2). The coating shall be furnished in accordance with standard commercial practice.

5.2 Instructions for use shall be furnished by the supplier.

6. NOTES

6.1 Intended use. The coating covered by this specification is intended for interior application on walls, ceilings, cabinets, furniture, and painted surfaces. The material has good washability. It can be used as a top coat for seamless flooring, where a gloss finish is not desired.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Administrative provisions for inspection records (see 4.1).
(c) Level of packaging and level of packing required (see 5.1).
(d) Size of container required (see 5.1).

CIVIL AGENCIES COORDINATING ACTIVITIES:

GSA-FSS
HUD-HEE

Preparing Activity: GSA-FSS

GSA DC 76-8196

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